



OPEN Multimodal sentiment analysis based on multi-layer feature fusion and multi-task learning

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Multimodal sentiment analysis (MSA) aims to use a variety of sensors to obtain and process information to predict the intensity and polarity of human emotions. The main challenges faced by current multi-modal sentiment analysis include: how the model extracts emotional information in a single modality and realizes the complementary transmission of multimodal information; how to output relatively stable predictions even when the sentiment embodied in a single modality is inconsistent with the multi-modal label; how can the model ensure high accuracy when a single modal information is incomplete or the feature extraction performance not good. Traditional methods do not take into account the interaction of unimodal contextual information and multi-modal information. They also ignore the independence and correlation of different modalities, which perform poorly when multimodal sentiment representations are asymmetric. To address these issues, this paper first proposes unimodal feature extraction network (UFEN) to extract unimodal features with stronger representation capabilities; then introduces multi-task fusion network (MTFN) to improve the correlation and fusion effect between multiple modalities. Multilayer feature extraction, attention mechanisms and Transformer are used in the model to mine potential relationships between features. Experimental results on MOSI, MOSEI and SIMS datasets show that the proposed method achieves better performance on multimodal sentiment analysis tasks compared with state-of-the-art baselines.

Multimodal sentiment analysis (MSA) has promoted the development of Human-Computer Interaction (HCI), education and medical care, public opinion analysis, point of interest recommendation¹ and other fields. Human sentiment has many forms of expression, such as text, facial expressions, speech, behavior, physiological signals, etc. Each type of information can be used as a modality for sentiment analysis. MSA aims to combine some modalities containing human sentiment information to achieve a more comprehensive and accurate perception. Since each modality has unique characteristics, fusing multiple modalities can making full use of the complementary and transfer characteristics of information. However, multimodal information also has heterogeneity. The semantics between different modal data are not completely consistent, as shown in Fig. 1. Simple modal addition cannot achieve the expected effect. This puts higher requirements on the feature extraction and fusion network of multimodal sentiment analysis.

MSA mainly includes two processes, Intra modal feature extraction and inter-modal information interaction. Intra modal feature extraction is to obtain a higher-dimensional sentiment representation, which focuses on the spatial or temporal information in a single modality. Features within a modality are the basis for improving the complementary characteristics of information between multiple modalities. Inter-modal information interaction focuses on the fusion of information. It makes full use of the transmission and complementary characteristics of information to provide more accurate and robust results for sentiment analysis tasks. These fusion methods are divided into various types according to the fusion method, including: data fusion, feature fusion, decision fusion, etc². Combining the above two processes, the model can learn the characteristics of the corresponding modalities and the interactions between different modalities.

Common methods for intra-modal feature extraction include recurrent neural networks such as LSTM and GRU. These methods can effectively capture sequence features. However, they cannot balance the relationship between intra-modal and inter-modal interactions, which results in that RNN alone cannot capture the multi-scale information expected for inter-modal interactions. The Tensor Fusion Network (TFN) proposed by Zedeh et al.³ uses LSTM to extract intra-modal feature information. And a triple Cartesian product is introduced to build a tensor fusion layer to simulate the dynamic fusion between different modal, although the computational complexity is high. Wang et al.⁴ combined GRU and Transformer to mine the intra-modal interactions, and

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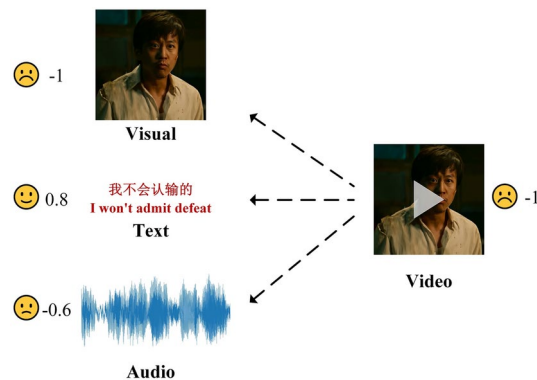


Fig. 1. The example of different modal sentiment representations.

integrated information into coding features to improve the correlation between modalities. For the inter-modal information interaction process, inspired by the attention mechanism, many models use attention-based strategies to fuse multi-modal features. Tsai et al.⁵ proposed Multimodal Transformer (MuT), which uses a multi-head attention mechanism to focus on features of different modalities and time series. The model can also produce stable predictions on unaligned data. However, it uses many fusion features for sentiment analysis but ignores the intra-modal features, which may lead to insufficient utilization of certain modal information. Sun et al.⁶ use the sentiment representation between modalities as context and interact with intra-modal features. This method better balances the relationship between intra-modal and inter-modal information interaction. Although numerous studies have proposed novel approaches for multimodal feature extraction and fusion, there is a lack of research focusing on precise sentiment analysis under conditions of modal imbalance and implicit expression, which are more prevalent in real-world scenarios.

Based on the above analysis, the main research motivation of this paper and the problem it aims to address are as follows: (1) Many intra-modal features cannot be fully extracted, directly fusion may easily ignore this unique information. (2) The complementarity of inter-modal features is not fully utilized, and the prediction results are more likely to be biased towards the semantic information represented by one modality. (3) In a complex semantic environment, the emotional tendencies of each modality are inconsistent, and it is difficult for the model to balance the relationship between intra-modal and inter-modal.

In this paper, we propose a Unimodal Feature Extraction Network (UFEN) to capture contextual information within a single modality. It combines multi-layer information fusion and self-attention mechanism based on the convolution process to obtain more important and unique features of intra-modal. It also reduces the impact of redundant information on subsequent inter-modal feature fusion. In the process of inter-modal information interaction, multi-head attention module can help the model learn more complementary information and high-dimensional representations between features. Based on this, we proposed a Multi-Task Fusion Network (MTFN), which is applied to the extraction of potential relationships between modalities and the overall sentiment representation of the fused features. This network introduces cross-modal attention module, Encoder and Decoder layer. In order to alleviate the problem of inconsistent sentiment representations in multiple modalities that reduce the accuracy of predictions, we set multi-modal sentiment analysis to multi-task learning. The feature representations of multiple tasks jointly affect the model, and the loss of each task is used to balance the relationship between intra-modal and inter-modal information interaction.

The main research objectives and contributions of this paper can be summarized as follows:

- Optimize and improve the quality and representational capacity of unimodal feature extraction: We propose a UFEN to extract single-modal semantic representation. This network is based on multi-layer information fusion, combining convolutional layers and self-attention modules. The network focuses on more representative information through time series features and different scale features;
- Maintain accurate information interaction and result prediction under conditions of sentiment asymmetry or implicit expression: We propose an MTFN, which treats MSA as a multi-task learning. This model can extract complementary features between modalities while alleviating the impact of asymmetry in inter-modal sentiment representation;
- The model can be applied to various sentiment analysis tasks and exhibits good generalization ability: To verify the effectiveness of the proposed method, we conduct comprehensive experiments on three widely used MSA benchmark datasets, including CMU-MOSI, CMU-MOSEI and CH-SIMS. The experimental results show that our method outperforms previous methods. The rest of this paper is structured as follows: “[Related works](#)” section introduces related work in recent years. In “[Methods](#)” section, we define the problem formulation and introduce the details of the method proposed in this paper. “[Experiments](#)” section describes the experimental dataset, evaluation metrics, experimental configurations and baseline models. “[Results](#)” section describes the experimental result analysis, ablation experiments and visualization analysis. Lastly, we draw our conclusions in “[Conclusion](#)” section.

Related works

In this section, we briefly review related work on multimodal sentiment analysis and multitask learning. In addition, we introduce how our work differs from previous work.

Multimodal sentiment analysis

Early sentiment analysis methods were unimodal and mainly focused on text modality.⁷ Text data alone cannot meet the demand for accurate analysis of more complex emotions in HCI. Therefore, multimodal emotion analysis⁸ had been proposed. In addition to text modality, multimodal sentiment analysis also includes speech and visual information, which can be used to predict implicit sentiment polarity more accurately. Research on multimodal sentiment analysis mainly focuses on representation learning and multimodal fusion⁹. For multi-modal representation learning, Hazarika et al.¹⁰ designed two different encoders to project features into modality-invariant subspaces and modality-specific subspaces respectively. Fang et al.¹¹ proposed a multi-modal similarity calculation network to learn embedding representations between text and images in a unified vector space. Zhang et al.¹² proposed an adversarial network to reduce the transfer of expression style in multi-modal cross-domain sentiment analysis and obtain joint representation between modalities. This approach facilitates the transfer of emotional information between different domains. Sun et al.¹³ proposed modality-invariant temporal learning techniques and inter-modal attention to obtain intra-modal and inter-modal features.

For multi-modal fusion, it can be mainly divided into early fusion and late fusion¹⁴. Early fusion usually refers to concatenating features from different modalities as input, while late fusion builds independent models for each modality and then integrates the output. Zadeh et al.⁵ fused multimodal features by external product of vectors. Yang et al.¹⁵ introduced attention to capture the representational relationships between different modalities before feature fusion. Zeng et al.¹⁶ used graph neural networks to fuse multi-source features and balance independent and complementary information between modalities. Li et al.¹⁷ used quantum theory to model the interaction between intra-modal and inter-modal, and used superposition and entanglement expressions at different stages. Majumder et al.¹⁸ adopt a hierarchical approach to fuse modal, first fusing unimodal, and then fusing all modal.

In addition, sentiment analysis research based on physiological signals such as electroencephalogram, electrocardiogram, and electrodermal activity signal are also popular. Kumar et al.¹⁹ used EEG and product review signals to perform sentiment analysis and obtain multi-modal ratings of consumer products. Katada et al.²⁰ combined electrodermal activity signal with linguistic representations to improve the effect of multi-modal sentiment analysis. At the same time, they used multiple unimodal models to prove that different physiological signals play different roles in sentiment analysis.

Deep learning method

Since deep learning inception, its methods have dominated the research in many directions. The field of affective computing, like other areas of computing research, has benefited greatly from advances in deep learning.

Daneshfar et al.²¹ explored the problem of model instability caused by different cultural and language differences. They used a transfer learning method based on training enhancement to achieve accurate cross-language sentiment analysis and achieved excellent results on the German sentiment analysis dataset. Zhang et al.²² addressed the issue of limited labeled speech data by using semi-supervised learning to generate pseudo-labels. They employed a knowledge transfer method to integrate cross-modal knowledge from image data into the learning process, achieving relatively accurate speech emotion analysis. Li et al.²³ aimed to address the issue of performance degradation in MSA caused by uncertain modality missing in practical applications. They proposed a related decoupled knowledge extraction method and used distillation learning algorithm to reconstruct the missing semantic information. The model achieved good results on the MOSI dataset. Meena et al.²⁴ applied the RNN method to develop a Point of Interest (POI) recommendation system. They achieved accurate sentiment analysis for multiple tasks including travel suggestions and group recommendations by integrating the SA model and the POI model. Paskaleva et al.²⁵ conducted research on interpretable emotion representation and expression generation, they developed a method that leverages labels of the non-unified models to annotate the novel unified one, and then used a diffusion model to generate images with a rich expression and additional attributes described by the conditioning text input.

Although deep learning methods have brought about huge performance improvements, there are still many aspects that need to be improved before meeting the growing expectations of users. The first is the interpretability of the model. Now more and more researchers are conducting research on this issue. Giving the model a complete physical meaning can make it easier to improve the model. The second is the model's ability to generalize outside the domain. Under the study of limited data, stronger generalization ability will enable the model to achieve better prediction results on different tasks and be closer to people's real needs for algorithms.

Multi-task learning

Multi-task learning aims to utilize the useful information contained in multiple related tasks to help improve the generalization performance of the model²⁶. Compared with single-task models, multi-task learning has three advantages: (1) By sharing information, the performance of the model can be improved; (2) It has better generalization ability; (3) Only one overall model is required, which reduces the model complexity. Multi-task learning has been widely used in MSA because of these advantages²⁷. Multi-task networks use parameter sharing methods to achieve the overall optimization of the model, which can be divided into hard parameter sharing and soft parameter sharing²⁸. Hard parameter sharing means that multiple tasks share several hidden layers of the network and make predictions for different tasks before output. Soft parameter sharing refers to using independent models for different tasks, as while as sharing and optimizing parameters between models.

Based on the interdependence of the two subtasks, Akhtar et al.²⁹ proposed a GRU-based multimodal attention framework for simultaneously predicting the sentiment and expressed emotions of an utterance. Foggia et al.²⁷ used multi-task convolutional neural networks to recognize gender, age, race and emotion from face images. At the same time, a weighted loss function is used to solve problems such as dataset imbalance and label loss. He et al.³⁰ proposed an interactive multi-task learning network that can perform joint learning of multiple related tasks at the token level and document level. The model utilizes a message passing architecture based on hard parameter sharing to pass shared latent variables to different tasks. Bai et al.³¹ proposed a multi-task extraction descent framework that iteratively optimizes a single cost function for each task through improved gradient descent. This method further improves the generalization performance of multi-task learning. Singh et al.³² introduced multi-task learning into complaint recognition and sentiment analysis, using deep neural networks and enhanced models based on common sense knowledge to obtain higher accuracy. Wang et al.³³ proposed a hierarchical multi-task learning framework for the end-to-end sentiment analysis task of simultaneously detecting aspect terms and sentiments, which utilizes the supervision of intermediate layers to fuse task-related features. Zhang et al.³⁴ proposed a broad transformer network, which uses features and fine-tuning methods to obtain high-level information of contextual representation.

In summary, previous methods tend to use fixed strategies to fuse the features of different modalities, without taking into account the differences between dimensions and modalities. The proposed multi-modal sentiment analysis model based on UFEN and MTFN can better mine the independent and complementary feature between intra-modal and inter-modal. Combining multi-layer information fusion with multi-task learning can improve the performance of sentiment analysis.

Methods

In this section, we describe the proposed method in detail. We first define the multimodal sentiment analysis task of this paper and explain the corresponding notation of the problem. Then we introduce the overall model of this article, including UFEN and MTFN. In addition, we also introduce the Transformer module we used. Finally, the multi-task learning method and overall optimization goal of the model is given.

Task definition

The MSA task in this paper refers to predicting the polarity and intensity of sentiment through video information. MSA task usually contains three main modalities: text (T), audio (A) and visual (V). We define the input of the model as $X_i \in \mathbb{R}^{(T_i \times D_i)}$, where T_i is sequence length and D_i denotes the feature dimension of modal i . Specifically, $i \in \{t, a, v\}$. The proposed model extracts the feature information of intra-modal through UFEN. Then MTFN is used for feature complementary fusion and multi-task learning. The output result of the UFEN is defined as Y_i , and the output of the MTFN is defined as Y_m and Y_m' . We take Y_m' as the final prediction result of the model in the multimodal sentiment analysis task, and use Y_i and Y_m to balance the information interaction of each task. The true label of data is defined as Y .

Model framework

The data used for multi-modal sentiment analysis can include easy-to-obtain external signals such as facial expressions, audio and text. It can also include physiological signals such as ECG, EEG, EMG and HR that are difficult to obtain but are more representative of real sentiment. The signals of these modalities have common characteristics that can reflect sentiment, as well as characteristics unique to a single modality. Utilizing the complementary characteristics of signals between different modalities can achieve more effective multi-modal sentiment analysis compared to single-modal sentiment analysis. Multimodal sentiment analysis has broader application prospects and meets people's needs for future HCI. In order to better balance the relationship between intra-modal feature extraction and inter-modal feature fusion, this paper proposes a model based on multi-layer feature fusion and multi-task learning, as shown in Fig. 2.

Firstly, we introduce a UFEN to extract intra-modal feature information and mine the interrelationships between multi-scale features. Assigns more weight to features that can express sentiment and reduces the adverse impact of redundant information on subsequent multi-modal feature fusion. Since multimodal sentiment analysis fuses multiple types of information and they contain different features for sentiment analysis, it is necessary to consider the hidden connections between different sensor data and assign different weights to these tasks. Based on this, in the second step, we construct an MTFN to strengthen information interaction between intra-model and inter-model. This network introduces cross-modal attention to combine representation information of different modalities. After connecting high-level features, we further capture feature information through the Encoder-Decoder layer of Transformer. In addition, we also introduce the idea of multi-task learning into multi-modal sentiment analysis, so that the model can adjust the importance of the modality according to the results and optimize the model.

Unimodal feature extraction network

UFEN is a multi-layer feature fusion model based on convolutional neural networks and self-attention for extracting intra-modal information. For the processing of heterogeneous data in different modalities, it uses multi-layer convolutional neural networks to obtain feature information of different dimensions, and then inputs the obtained multi-scale information into the self-attention module to adjust the weights of different features. Adding Bi-GRU and Multi-layer feature fusion enables the model to obtain temporal features based on the contextual information of the data. By fusing this information, it helps to improve the model's representation ability and comprehensive perception of global and local information. For intra-model feature extraction, UFEN can meet the feature extraction needs of different scale data by changing the number of convolution layers, which

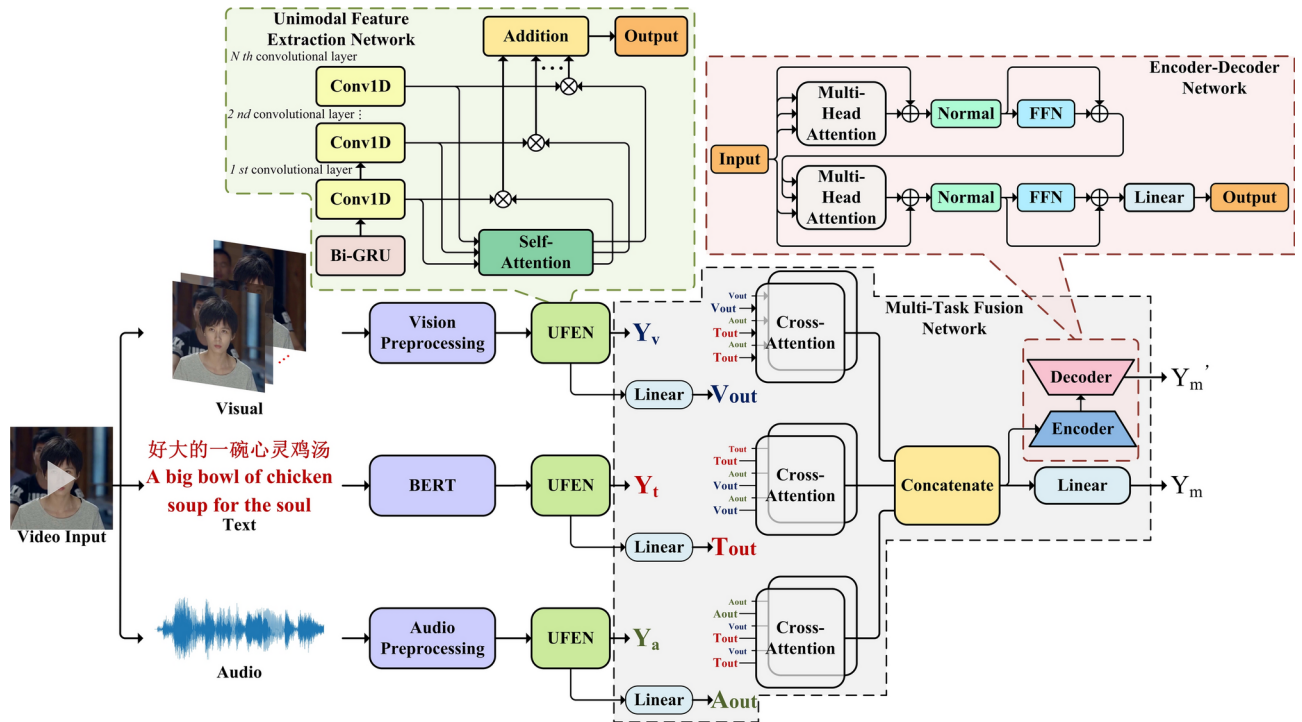


Fig. 2. The overall architecture of our method. The Y_m , Y_t , Y_a , Y_v , Y_m' are the prediction outputs of the multi-modal task, the three single-modal tasks and the fusion feature reconstruction task respectively. Y_m' is used as the multimodal sentiment output.

can obtain more fine-grained features. UFEN has good generalization performance to adapt to the needs of different tasks, the structure of UFEN is shown in Fig. 3.

For different modality, we use different methods to preprocess and then input to UFEN. The input is defined as $X_i \in \mathbb{R}^{(T_i \times D_i)}$, where $i \in \{t, a, v\}$ is used to represent the current mode. Bi-GRU is used to obtain the temporal characteristics and contextual sentiment information of data. In order to make the model pay attention to global features and local features at the same time, we introduce convolution and self-attention for multi-layer feature fusion. Data preprocessing can be described as:

$$X_i = \text{Perprocess}(X_i) \in \mathbb{R}^{T_i \times D_i} \quad (1)$$

Based on the classic GRU network architecture, the Bi-GRU layer simultaneously extracts the hidden state between information in the forward and backward propagation processes. Bi-GRU can be expressed as:

$$\begin{cases} \vec{h}_t = \overrightarrow{GRU}\left(X_t, \vec{h}_{t-1}\right) \\ \overleftarrow{h}_t = \overleftarrow{GRU}\left(X_t, \overleftarrow{h}_{t-1}\right) \\ h_t = [\vec{h}_t, \overleftarrow{h}_t] \end{cases} \quad (2)$$

Where X_t represents the input information of the current time step. h_{t-1} is the GRU output of the previous time step. GRU includes the calculation of parameters such as reset gates, update gates, and hidden states. The forward hidden representation $\vec{h}_t$ and the backward hidden representation $\overleftarrow{h}_t$ are spliced to obtain h_t . The final output of Bi-GRU is expressed as X_i^{Bi-GRU} , which is in the low-dimensional space $\mathbb{R}^{d_m}$.

$$X_i^{Bi-GRU} = BiGRU(X_i) \quad (3)$$

After obtaining time series information, the model focuses on multi-dimensional information extraction and fusion. Therefore, multi-layer convolutional feature extraction process is introduced, which can be described as:

$$X_i^{layer} = Conv1D^{layer}(X_i^{Bi-GRU}, K, F) \quad (4)$$

Where $layer$ represents the number of one-dimensional convolution layers, K is the kernel size of the convolution layer, and F is the number of filters in the convolution layer. For $layer \in \{1, 2, \dots, l\}$, $F \in \{F_1, F_2, \dots, F_l\}$ and $X_i^{layer} \in \{X_i^1, X_i^2, \dots, X_i^l\}$ represents the output of different number convolutional layers.

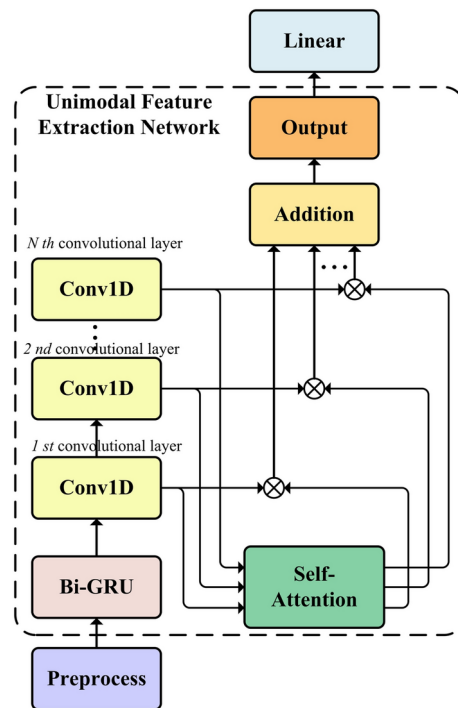


Fig. 3. Illustration of the UFEN.

In order to reduce redundant information and integrate multi-level features, we combine convolution with attention weights. The attention mechanism is used to highlight the most relevant features from the unimodal feature extraction process, allowing the model to focus on both local and global contextual information. It can be described as:

$$\begin{cases} X_i^{layerAtt} = Attention(X_i^{layer}) \\ X_i^{self-att} = X_i^{layer} \otimes X_i^{layerAtt} \end{cases} \quad (5)$$

Where $Attention(\cdot)$ represents attention calculation. When Q , K , and V are all in the same mode, it can be regarded as self-attention. The detailed process of the attention mechanism is shown in “Cross-modal attention”.

In order to ensure that the features enhanced by attention are aligned with the dimensions required for subsequent processing, an up-sampling operation is performed after the attention mechanism, can be described as:

$$X_{layer} = Unpooling(X_i^{self-att}) \in \mathbb{R}^{d_m} \quad (6)$$

Where $X_i^{fusion} \in \mathbb{R}^{d_m}$ is the output of UFEN. The up-sampled features of each layer are fused by matrix addition, which is expressed as:

$$X_i^{fusion} = X_1 \oplus X_2 \oplus \dots \oplus X_{layer} \quad (7)$$

Where $\oplus$ stands for element addition. This method can not only retain the fine-grained information extracted by different convolutional layers, but also integrate the global dependency information captured by the attention mechanism.

In order to meet the needs of subsequent multi-task learning, we add a linear layer to implement single-modal sentiment analysis. The specific process is:

$$Y_i = softmax(W_i X_i^{fusion} + b_i) \quad (8)$$

Where X_i^{fusion} is the hidden state, which is used for inter-modal feature fusion, W_i is the weight matrix, and b_i is the corresponding bias. Y_i is a single-modal label, used for multi-task learning.

Multi-task fusion network

For heterogeneous data in sentiment analysis task, there are both low-level features that can represent specific information of a certain modality, and high-dimensional features with global semantic information of multiple modalities. It is difficult for the model to strike a balance between intra-modal feature extraction and inter-modal information fusion. In this case, using multi-task learning method to give higher weight to the dominant modal can improve the stability of the sentiment analysis results. In order to improve the effectiveness of multi-modal fusion, this paper designs an MTFN model to extract inter-modal features by exploring multi-task learning and information interaction between modalities. MTFN mainly consists of two parts: (1) Cross-modal multi-head attention (“Cross-modal attention”) used to fuse different modal features; (2) Encoder–Decoder layer (“Encoder–decoder layer”) used to enhance feature extraction and information representation of multi-task learning; (3) Multi-Task learning used to combat data imbalance and improve generalization performance (“Multi-task learning”).

Cross-modal attention

We adopt cross-modal multi-head attention to enhance information interaction between modalities. In addition, fuse global features and local features to obtain more representations. For modalities i and j , we define modal i as Query and calculate the mutual relationship between i and j . The cross-modal attention from i to j can be expressed as $i \rightarrow j$, and the structure is as shown in Fig. 4.

First, we use linear to map intra-feature of modal i and modal j , $X_i = W_i X_i + b_i$ and $X_j = W_j X_j + b_j$, where $W_i \in \mathbb{R}^{D_e \times D_i}$, $W_j \in \mathbb{R}^{D_e \times D_j}$. Then the modal i in the two-dimensional space is defined as Query, and the modal j is defined as Key and Value, and the specific process is as follows:

$$\begin{cases} Q_i = W_{Q_i} X_i \\ K_j = W_{K_j} X_j \\ V_j = W_{V_j} X_j \end{cases} \quad (9)$$

Where $Q_i \in \mathbb{R}^{D_e \times D_i}$ and $K_j, V_j \in \mathbb{R}^{D_e \times D_j}$. Cross-modal representation can be obtained by calculating the similarity between Query and Key, then weighted summation with Value.

$$CrossAtt(Q_i, K_j, V_j) = softmax\left(\frac{Q_i K_j^T}{\sqrt{d_k}}\right) V_j \quad (10)$$

Where d_k is the set dimension of Query and Key. Different from single-head cross-modal representation, multi-head cross-modal attention uses different parameters to perform multiple linear projections on the Query, Key, and Value. It can represent the features of different subspaces cross-modally. Multiple features are concatenated and linearly projected again to obtain the final result. The specific process is as follows:

$$\begin{cases} head_n = CrossAtt(Q_{i,n}, K_{j,n}, V_{j,n}) \\ MhAtt(Q_i, K_j, V_j) = Concat(head_1, \dots, head_n) W \end{cases} \quad (11)$$

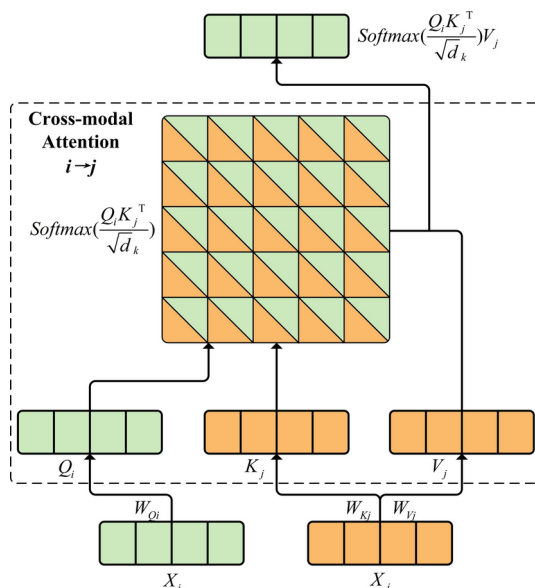


Fig. 4. Illustration of cross-modal attention.

Where head is the cross-modal attention output in the subspace, $\text{head}_n \in \mathbb{R}^{\frac{D_e}{n}}$, $Q_{i,n} \in \mathbb{R}^{\frac{D_e \times D_i}{n}}$, $K_{j,n}, V_{j,n} \in \mathbb{R}^{\frac{D_e \times D_j}{n}}$ and $W \in \mathbb{R}^{D_e \times D_e}$. $MhAtt(\cdot)$ represents multi-head attention. After cross-modal multi-head attention processing, the output results are layer normalized to obtain the inter-modal features of i and j , expressed as:

$$\text{Output}_{\text{CrossAtt}} = \text{LN} (MhAtt(Q_i, K_j, V_j)) \quad (12)$$

Encoder-decoder layer

After intra-modal and inter-modal feature extraction, the model can obtain multi-modal fusion features. In order to enhance the accuracy of the model in extracting sentiment features from multimodal data, we constructed an Encoder-Decoder layer, as shown in Fig. 5.

The higher-dimensional features of the fused modality are obtained through the encoder network, and then the decoder network is used to reconstruct the features. We incorporate the labels predicted by the encoder-decoder network and the labels predicted based on the fusion features into multi-task learning, which can improve the generalization performance of the model.

The Encoder-Decoder network contains three sub-network layers, including multi-head attention mechanism, layer normalization and feedforward network. Among them, the input of the multi-head attention layer in the decoder increases the output of the encoder.

First, the results $CA_{i \rightarrow i'}$ of cross-modal attention between different modalities are concatenated as the input of Encoder-Decoder. The specific process can be expressed as:

$$\text{Input} = \text{Concat}(CA_{i \rightarrow i'}), \quad i \in t, a, v \quad (13)$$

Where $\text{Input} \in \mathbb{R}^{d_b \times 6d_m}$, d_b is batchsize. d_m is the single-modal feature dimension. For a given input, the first sub-layers of encoder and decoder are multi-head attention and layer normalization. The output of layer normal can be summarized as:

$$\begin{cases} Nm_{en} = \text{LN}(\text{Input} + MhAtt(\text{Input})) \\ Nm_{de} = \text{LN}(\text{Input} + MhAtt(\text{Input}, \text{Output}_{en})) \end{cases} \quad (14)$$

Where Nm, en, de represents normal, encoder and decoder. $MhAtt(\cdot)$ in the formula is introduced in “Cross-modal attention”. The output calculation process of decoder and encoder are:

$$\begin{cases} \text{Output}_{en} = Nm_{en} + FF(Nm_{en}) \\ \text{Output}_{de} = Nm_{de} + FF(Nm_{de}) \end{cases} \quad (15)$$

Where $FF(\cdot)$ represents feed forward network. In addition, we add linear layers to the input and output signals of encoder-decoder for subsequent multi-task learning. The purpose of adding a linear layer after the decoder is to evaluate the reconstructed feature results and improve the generalization performance of the network. Introducing the predicted labels of reconstructed sentiment information into multi-task learning can optimize model parameters. This approach can also prevent the model from predicting high-dimensional information into a fixed label due to imbalance training or uneven sentiment representation of data. These two Linear layers can be expressed as:

$$\begin{cases} Y_m' = \text{softmax}(W_{de}\text{Output}_{de} + b_{de}) \\ Y_m = \text{softmax}(W_m\text{Input} + b_m) \end{cases} \quad (16)$$

Where W_{de} and W_m are weight matrices, b_{de} and b_m are the corresponding biases. Y_m' and Y_m represent the prediction labels.

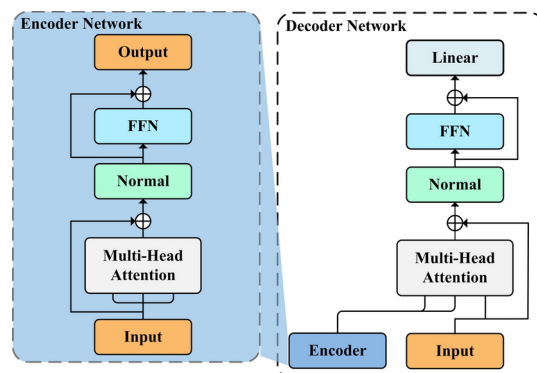


Fig. 5. Illustration of encoder-decoder layer.

Multi-task learning

Video, audio and text information are highly relevant to human sentiment representation. Common sentiment analysis methods include single-modal recognition for a certain type of information and multi-modal recognition for fused features. These methods have limitations in result accuracy and fine-grained sentiment feature extraction. Focusing only on a single task may ignore some related tasks that have potential information to improve model performance. Therefore, considering single-modality and fused-modality prediction as sub-tasks. By sharing representations, the complementary properties of multimodal sentiment representations can be better exploited. The generalization performance of the model will also improve. At the same time, the use of multi-task learning means that only one model is needed to handle the independent optimization and joint optimization of multiple modalities. Compared with the fusion of multiple models, multi-task learning can reduce the computational complexity of the model.

By combining the single-modal subtask, the multi-modal fusion feature subtask and the multi-modal feature reconstruction subtask, the total loss $\mathcal{L}_{total}$ can be expressed as:

$$\mathcal{L}_{total} = \sum_{i \in \{t, a, v\}} \mathcal{L}_{gt}^i + \mathcal{L}_{gt}^f + \mathcal{L}_{gt}^{en-de} \quad (17)$$

Where $\mathcal{L}_{gt}^i$ represents the loss between the single-modal prediction result Y_i and the ground truth Y . $\mathcal{L}_{gt}^f$ and $\mathcal{L}_{gt}^{en-de}$ respectively represent the loss between multi-modal fusion prediction Y_m , encoder-decoder layer prediction Y_m' and the ground truth Y . All loss function calculations in the model use MSE (Mean Squared Error), which can be expressed as:

$$\mathcal{L}_{gt}^x = \frac{1}{N} \sum_{n=1}^N (y^n - \hat{y}^n)^2 \quad (18)$$

Where $\mathcal{L}_{gt}^x$ represents the loss between the prediction results $\hat{y}^n$ of different subtasks and the true value y^n , and N is the number of training samples. The pseudocode for the entire training pipeline is illustrated in Algorithm 1.

Input: unimodal features X_i , $i \in \{t, a, v\}$ with the corresponding ground truth labels Y

Output: *model*

```

1: Initialize model parameters
2: for each sample in the mini-batch do
3:   (1) UFEN (Uni-modal Feature Extraction Network)
4:   for each modality  $i \in \{t, a, v\}$  do
5:     Obtain the preprocessed input features  $X_i$  according to Eq.1
6:     Calculate the output  $X_i^{Bi-GRU}$  of Bi-GRU layer according to Eq.2 and Eq.3
7:     Extract multi-layer sentiment features  $X_i^{layer}$  according to Eq.4
8:     Calculate feature weights and up-sample features to the same size
9:     Extract multi-layer fusion features  $X_i^{fusion}$  according to Eq.5
10:    Obtain single-modal prediction result  $Y_i$  as in Eq.6
11:    Calculate the loss  $\mathcal{L}_{gt}^i$  between  $Y_i$  and ground truth  $Y$ 
12:  end for
13:  (2) MTFN (Multi-Task Fusion Network)
14:  for each modality  $i \in \{t, a, v\}$  do
15:    Compute cross-modal representations  $CA_{i \rightarrow i'}$  between different modalities according to Eq.10
16:    Compute multi-modal fusion features  $CA_M$  according to Eq.11
17:    Obtain multi-modal prediction result  $Y_m$  as in Eq.14
18:    Calculate the loss  $\mathcal{L}_{gt}^f$  between  $Y_m$  and ground truth  $Y$ 
19:    Calculate the reconstruction features  $Output_{de}$  of the encoder-decoder layer according to Eq.13
20:    Obtain the prediction results  $Y_m'$  of encoder-decoder layer as in Eq.14
21:    Calculate the loss  $\mathcal{L}_{gt}^{en-de}$  between  $Y_m'$  and ground truth  $Y$ 
22:  end for
23:  (3) Total Optimization
24:  Compute total training loss  $\mathcal{L}_{total}$  as in Eq.15
25:  Update model parameters
26: end for
27: return model

```

Algorithm 1. Training algorithm.

Experiments

In this section, we introduce the datasets used in the experiment, detailed parameter settings, reference baselines and evaluation methods. The experiment involves three modalities: video, text, and audio.

Dataset

CH-SIMS: The dataset CH-SIMS³⁵ contains 2281 carefully edited Chinese video clips from 60 film and television videos. The video clips have rich character backgrounds and a wide age range. The voice part of the video is in Mandarin, the length of the short clip is no less than 1 second and no more than 10 seconds. For each video clip, no other faces appear except the face of the speaker. The annotation of the dataset isolates information from other irrelevant modalities. The annotation order is text first, audio second, silent video third, and multimodal last. The annotation task includes 2 categories (Positive, Negative), 3 categories (Positive, Negative, Neutral) and 5 categories (Negative, weakly Negative, neutral, weakly positive, Positive). This dataset has accurate multimodal and independent unimodal annotations, which can be used to support researchers in multimodal or unimodal sentiment analysis.

MOSI: The Multimodal Corpus of Sentiment Intensity (MOSI)³⁶ dataset aims to highlight three characteristics: the diversity of viewpoint information, the importance of emotional intensity, and the complementarity between modalities. The author collected 2199 video clips through online sharing website, ranging in length from 2 to 5 min. The dataset contains video, speech, and text data. Each video clip is labeled with an intensity label of -3 (strongly negative) to 3 (strongly positive).

MOSEI: The Multimodal Opinion Sentiment and Emotion Intensity (MOSEI)³⁷ dataset is aimed at improving the diversity in the training samples, the variety in the topics and annotations of the current dataset. The authors collected 23,453 annotated video clips through online video sharing websites. The video clips come from 1000 different speakers and 250 topics. Each video segment contains manual transcription aligned with audio to phoneme level. The author also used a variety of methods to annotate the data, including a $[-3, 3]$ Likert scale of: $[-3$ highly negative, -2 negative, -1 weakly negative, 0 neutral, $+1$ weakly positive, $+2$ positive, $+3$ highly positive] and a scale of $[0$ no evidence, 1 weak, 2 mid, 3 high] of Ekman emotions. The detailed statistics of SIMS, MOSI and MOSEI datasets are shown in Table 1.

Experiments settings

In this article, our model training is based on NVIDIA GeForce RTX 3060Ti GPU and python3.9 + pytorch1.12.1 + cuda11.6 deep learning framework. In the experiment, we use the Adam optimizer to optimize the model parameters. For the MOSI, MOSEI and SIMS datasets we set three sets of hyperparameters, as shown in Table 2.

In order to make a fair comparison with other baselines, we follow the preprocessing method of the previous work^{38–40} to extract features of images, audios and texts. For visual information, we use Facet⁴¹ to extract facial expression features of MOSI and MOSEI datasets, and use Openface2.0 toolkit⁴² to extract features of SIMS dataset. Facet first identifies the face in the video image frame and key landmarks on the face, such as eyebrows, the tip of nose, the corners of mouth. Then feature information including action units, facial landmarks and interocular distance. The action units contain 20 features. Openface2.0 toolkit is similar to Facet. It can also perform facial landmark detection, head pose estimation, facial action unit recognition and eye-gaze estimation, and the extracted features are spliced in sequence. For speech information, we use the Librosa⁴³ speech toolkit to extract the acoustic features of the SIMS dataset, and use the COVAREP⁴⁴ toolkit to extract the acoustic features of the MOSI and MOSEI datasets. Librosa is a commonly used Python package for audio signal processing. It is used to extract frame-level acoustic features of SIMS data at 22,050 Hz, with a total of 33 dimensions. Features include 1-dimensional logarithmic fundamental frequency (log F0), 20-dimensional Mel frequency cepstral coefficients (MFCCs) and 12 dimensional Constant-Q chromagram (CQT), which are closely related to emotions and tone of speech. COVAREP software is similar to Librosa and is used to extract acoustic features related to emotion and intonation, including 12 Mel-frequency cepstral coefficients, pitch, voiced/unvoiced segmenting features, glottal source parameters, peak slope parameters and maxima dispersion quotients. For text modality, we adopt the BERT⁴⁵ pre-trained model as the feature extractor. BERT aims to learn word embeddings by pretraining deep bidirectional representations from unlabeled text, which jointly condition on both left and right contexts across all layers. Different BERT pre-training models are used for English and Chinese texts. Due to the characteristics of BERT, there is no need to use word segmentation tools before feature extraction. After preprocessing, all modal features are arranged in the order of shape[batch size, sequence length, feature dimensions].

Baselines and experimental metrics

In order to verify the advantages of our method, we compare the performance of the proposed method with the MSN baselines. The details of these baselines are as follows:

EF-LSTM⁴⁶: Early Fusion LSTM (EF-LSTM) utilizes feature-level fusion and sequence learning of Bidirectional Long-Short Term Memory (Bi-LSTM) deep neural networks.

LF-DNN⁴⁷: Late fusion Deep Neural Network (LF-DNN) introduces three model structures to encode multimodal data, and then combines PCA for early feature fusion and late decision fusion.

TFN³: Tensor Fusion Network (TFN) introduces an end-to-end fusion method for sentiment analysis. The tensor fusion layer explicitly models the unimodal, bimodal and trimodal interactions using a 3-fold Cartesian product from modality embeddings.

LMF⁴⁸: Low-rank Multimodal Fusion (LMF) utilizes low-rank factors for multimodal representation and making multimodal feature fusion more efficient.

MulT⁵: Multimodal Transformer (MulT) utilizes a bidirectional cross-modal attention mechanism to focus on the interactions between multi-modal data at different time steps.

Datasets	Train	Valid	Test	Total	Range of labels
SIMS	1368	456	457	2281	$[-1, 1]$
MOSI	1284	229	686	2199	$[-3, 3]$
MOSEI	16326	1871	4659	22865	$[-3, 3]$

Table 1. The detailed statistics of SIMS, MOSI and MOSEI datasets.

Hyper-parameter	MOSI	MOSEI	SIMS
Learning rate	5e-3	1e-4	5e-3
Batch size	32	32	16
Self-Att head	1	1	1
Cross-Att head	4	4	4
Att dropout	0.2	0.2	0.2
Dropout	0.1	0.1	0.1
Early stop	8	8	8
Epoch	30	50	50

Table 2. Model hyper-parameters.

MISA¹⁰: Modality-Invariant and Specific Representations(MISA) translates each modality into two different subspaces, learn commonalities and reduce modal gaps through modality-invariant subspaces, then capture private features through modality-specific subspaces.

Self-MM⁴⁹: Self-Supervised Multi-task Multimodal (Self-MM) sentiment analysis network designs a label generation module based on the self-supervised learning strategy to acquire independent unimodal supervisions.

MFN⁵⁰: Memory Fusion Network (MFN) proposes a Delta-Memory Attention Network (DMAN) to identify cross-view interactions while summarizing through Multi-view Gated Memory.

Graph-MFN³⁷: This model introduces the Dynamic Fusion Graph (DFG) module based on the MFN network and performs fusion analysis.

MMIM⁵¹: Multimodal Mutual Information Maximization (MMIM) reduces task-related information loss by increasing the mutual information between unimodal representations, hybrid results and low-level unimodal representations.

In this article, we use two evaluation indicators, classification and regression, to evaluate the model. On MOSI and MOSEI, following previous works^{52–54}, our evaluation indicators include binary accuracy (Acc-2), seven-class accuracy (Acc-7), F1-score, mean absolute error (MAE) and Pearson correlation coefficient (Corr). It is worth noting that Acc-2 have two different representation methods, namely negative/non-negative³⁷ and negative/positive. Therefore, our results give two representations and use segmentation markers -/- to distinguish them, where the left score is negative/non-negative and the right score is negative/positive. On SIMS, following previous works^{35,55}, our evaluation indicators include binary accuracy (Acc-2), triple accuracy (Acc-3), five-class accuracy (Acc-5), F1-score, MAE and Corr.

Acc- x is used to evaluate whether the model can divide the data into corresponding sentiment intervals, expressed as:

$$Acc_x = \frac{1}{N} \sum_{n=1}^N \mathbb{I}(\hat{y}^n \in I(y^n)) \quad (19)$$

N represents the number of samples involved in model evaluation, $\mathbb{I}()$ is an indicator function. When the model prediction value $\hat{y}^n$ is within the sentiment interval of ground truth, it outputs 1, otherwise 0. x represents the number of sentiment classifications. As x increases, there are more sentiment intervals $\mathbb{I}()$, and the sentiment analysis becomes more detailed. Acc_x can also be calculated through the confusion matrix, which is expressed as:

$$Acc_x = \frac{TP + TN}{TP + TN + FP + FN} \quad (20)$$

F1-score is used to evaluate the performance of the model under data imbalance conditions. It combines precision Pc and recall Rc , can be expressed as:

$$\begin{cases} F1score = 2 \times \frac{Pc \cdot Rc}{Pc + Rc} \\ Pc = \frac{TP}{TP + FP} \\ Rc = \frac{TP}{TP + FN} \end{cases} \quad (21)$$

MAE is used to measure the accuracy of model regression and is expressed as:

$$MAE = \frac{1}{N} \sum_{n=1}^N |y^n - \hat{y}^n| \quad (22)$$

y^n and $\hat{y}^n$ represent the true label of data and the predicted label of the model respectively. For all metrics except MAE, higher values indicate better performance. For all metrics except MAE, higher values indicate better performance.

Results

In this section, we compare the proposed model with baselines on SIMS, MOSI and MOSEI datasets. We conduct ablation experiments and visual analysis simultaneously to demonstrate the effectiveness of the proposed method.

Comparisons to baselines

In this subsection, we compare the proposed model with baselines for multimodal sentiment analysis. We perform model evaluation on the English dataset CMU-MOSI, CMU-MOSEI and the Chinese dataset CH-SIMS. The experimental results are shown in Tables 3, 4 and 5.

For the CMU-MOSI dataset, we can see the results from Table 3. Self-supervised learning-based self-MM and modal feature projection-based MISA outperform other baseline methods. However, our method based on multi-layer feature fusion and multi-task learning outperforms MISA and Self-MM on multiple evaluation metrics. Specifically, our method outperforms Self-mm by 2.7%/2.8% and outperforms MISA by 3.4%/3.2% on Acc2. The method is 2.7%/2.8%, 0.03, and 0.013 better than Self-mm in F1-score, MAE, and Correlation in bipolar sentiment analysis. At the same time, it is 3.5%/3.1%, 0.055, and 0.031 better than MISA. For Acc7, the method is 1.8% and 4.4% higher than Self-mm and MISA.

In order to better evaluate the performance of the model on MOSI dataset, we visualized the results of the two-class and seven-class classification of sentiment polarity. The confusion matrix is shown in Fig. 6. For the classification of two polarities, our method maintains a high classification accuracy. For the classification of seven sentiment polarities, our method performs better on test samples with obvious sentiment polarity. For weak sentiment polarity samples, although the classification results outperform baseline models, there is still risk of misclassifying them as neutral. This situation is more obvious in weak negative samples. We conclude that this may be due to the limited number of features that express weak emotions. When the sentiment features of each modality are inconsistent, the model is more inclined to neutral prediction results.

For the CMU-MOSEI dataset, we can see the results from the Table 4. Due to the increase in data volume and sentiment information, the performance of all baselines for multipolar sentiment analysis has improved. MMIM based on mutual information maximization and MISA based on modal feature projection outperform other baseline methods. Compared with the experimental results of all baselines on the MOSEI dataset, our method has improved to varying degrees. Specifically, our method outperforms MMIM by 2.7%/2.2% and outperforms MISA by 1.2%/0.5% on Acc2. The method is 2.6%/2.2%, 0.064 and 0.02 better than MMIM in F1-score, MAE and Correlation in bipolar sentiment analysis. At the same time, it is 1.1%/0.5%, 0.02, and 0.004 better than MISA. For Acc7, the method is 2.6% and 2.3% higher compared to MMIM and MISA. Our method has better performance on complex tasks.

For the CH-SIMS dataset, our method improves on most evaluation indicators, as shown in Table 5. It is 1.91% higher than Graph-MFN based on Dynamic Fusion Graph and MulT based on cross-modal attention on Acc2. It is 0.87 and 0.61 higher than Graph-MFN and tensor fusion-based TFN on F1-score. The model is close to the best-performing LF-DNN on MAE, with a difference of 0.004. And on Correlation, it is 0.012 different from Self-mm. The model improves both Acc3 and Acc5, with the highest improvements reaching 16.35% and 24.41%. The smallest improvements are 0.37% and 1.66%.

Judging from the experimental results of the three datasets, MFN performs poorly on CMU-MOSI and MOSEI, especially on Acc2, F1-score and Correlation. The reason may be that although MFN takes into account the intra-modal and inter-modal correlation of features, only using LSTM may not be able to fully extract the sentiment information in the data. At the same time, using all single-modal information for feature fusion will also lead to a reduction in SNR. EF-LSTM has the worst performance on the CH-SIMS dataset, mainly reflected in Acc2, F1-Score and Correlation. This may be due to the fact that early fusion is more used for coarse-grained feature extraction and the fused features are strongly affected by noise of each modality. This makes it difficult for fused features to accurately represent the connections between modalities.

In order to better evaluate the performance of the model on SIMS dataset, we visualized the results of the three-class and five-class classification of sentiment polarity. The confusion matrix is shown in Fig. 7. For the classification of three polarities, our method maintains a high classification accuracy. For the classification of five sentiment polarities, the results are similar to MOSI, the model has a chance of misclassifying other sentiment polarities as neutral, but its performance is better than that of MOSI. We conclude that this may be due to the SIMS dataset has more sentiment features.

To evaluate the novelty of our method in multimodal sentiment analysis research, we collected recent works from other research areas for comparison. The results are shown in Table 6.

As shown in the table, the method proposed in this paper has also gained attention from researchers in other fields and has demonstrated good performance across various tasks.

Models	MOSI				Acc7
	Acc2	F1-score	MAE	Corr	
EF-LSTM ^a	77.8/79.0	77.7/78.9	0.952	0.651	34.5
LF-DNN ^a	78.0/79.3	77.9/79.3	0.978	0.658	33.6
TFN ^b	-/80.8	-/80.7	0.901	0.698	32.1
LMF ^b	-/82.5	-/82.4	0.917	0.695	32.8
MFN ^b	77.4/-	77.3/-	0.965	0.632	34.1
Graph-MFN ^a	77.9/80.2	77.8/80.1	0.939	0.656	34.4
MuT ^b	81.5/84.1	80.6/83.9	0.861	0.711	40.0
MISA ^b	81.8/83.4	81.7/83.6	0.783	0.761	42.3
Self-MM*	82.5/83.8	82.5/83.9	0.758	0.779	44.9
MMIM*	80.2/81.9	80.2/82.0	0.811	0.725	45.0
Our Method	85.2/86.6	85.2/86.7	0.728	0.792	46.7

Table 3. Performance comparison between our method and baselines on MOSI Dataset. ^a Is from⁵³. ^b Is from¹⁰. Model with * are reproduced from the open-source code provided in original papers. For all metrics except MAE, higher values indicate better performance. Significant values are in bold.

Models	MOSEI				Acc7
	Acc2	F1-score	MAE	Corr	
EF-LSTM ^a	80.1/80.3	80.3/81.0	0.603	0.682	49.3
LF-DNN ^a	78.6/82.3	79.0/82.2	0.561	0.723	52.1
TFN ^b	-/82.5	-/82.1	0.593	0.700	50.2
LMF ^b	-/82.0	-/82.1	0.623	0.677	48.0
MFN ^b	76.0/-	76.0/-	-	-	-
Graph-MFN ^a	81.9/84.0	82.1/83.8	0.569	0.725	51.9
MuT ^b	-/82.5	-/82.3	0.58	0.703	51.8
MISA ^b	83.6/85.5	83.8/85.3	0.555	0.756	52.2
Self-MM*	77.6/83.9	78.5/84.0	0.540	0.757	54.2
MMIM*	82.1/83.8	82.3/83.6	0.599	0.740	51.9
Our Method	84.8/86.0	84.9/85.8	0.535	0.760	54.5

Table 4. Performance comparison between our method and baselines on MOSEI Dataset. ^a Is from⁵³. ^b Is from¹⁰. Model with * are reproduced from the open-source code provided in original papers. For all metrics except MAE, higher values indicate better performance. Significant values are in bold.

Models	SIMS				Acc3	Acc5
	Acc2	F1-score	MAE	Corr		
EF-LSTM ^a	69.37	56.82	0.599	0.521	54.27	21.23
LF-DNN ^a	78.99	79.72	0.419	0.589	64.99	41.36
TFN ^a	80.31	80.66	0.451	0.581	64.33	36.76
LMF ^a	78.99	78.99	0.442	0.574	66.74	37.86
MFN ^a	79.21	79.15	0.434	0.581	66.08	39.39
Graph-MFN ^a	79.65	80.40	0.477	0.581	67.4	39.17
MuT ^a	79.65	79.94	0.439	0.582	65.86	38.95
MISA ^b	69.37	56.82	0.587	0.113	51.42	20.79
Self-MM*	78.77	78.88	0.425	0.591	63.68	39.82
MMIM*	73.96	74.59	0.441	0.531	60.61	43.54
Our method	81.56	81.27	0.423	0.583	67.77	45.20

Table 5. Performance comparison between our method and baselines on SIMS Dataset. ^a Is from³⁹. ^b Is from⁵². Model with * are reproduced from the open-source code provided in original papers. For all metrics except MAE, higher values indicate better performance. Significant values are in bold.

Ablation study

To further observe the performance of the model and the importance of different modalities, we conduct a series of ablation experiments on the SIMS and MOSI datasets in this section. These are used to study the effectiveness of each component in the multi-layer feature extraction and multi-task fusion network.

The experimental results in Table 7 show the impact of the number of CNN and self-attention layers in UFEN. It can be seen from the results that the effects of multi-layer feature extraction are better than that of single-layer feature extraction. However, as the number of layers increases, the experimental results show a trend of first getting better and then gradually getting worse. When the number of feature extraction layers is set to 2, the model performs best. This is due to the model giving more weight to the temporal sentiment representation information in single-modal data. Through two layers of intra-modal information extraction, the SNR of single-modal features is improved, which lays a good foundation for subsequent multi-modal fusion. When the number of UFEN layers increases to 3 and 4, the performance of the model gradually becomes worse. The reason may be that over-extraction leads to an increase in the imbalance of sentiment information after feature fusion, and the model tends to overfit. At the same time, deeper feature extraction requires a larger amount of training data.

Table 8 shows the sentiment analysis results of fused modality combine with different single-modality tasks.

It can be seen from the results that the introduction of unimodal subtasks can significantly improve the performance of the model. With the number of subtasks increase, the model tends to perform better. Taking all the results together, the gain brought by the visual modality is the most obvious. This is because the visual modality has a large amount of image information, including the movements of different facial areas and micro-expressions. The text mode provides more gain than the audio mode. We analyze that this is because compared to speech, text has been abstracted and compressed by humans, which is obviously more accurate than the speech features extracted by the model. At the same time, sentence vocabulary in text modality already contains rich semantic knowledge. However, due to differences in personalities or cultural influences of different people, it is more difficult to accurately extract sentiment representations in audio modality.

To better understand the contributions of different modules in the model, we conduct ablation studies on UFEN and MTFN, as shown in Table 9.

Ablation on UFEN

(1) Role of self-attention layer and Conv1D

After removing the self-attention and Conv1D layer respectively, the performance of all indicators decreased. This is because the single-modal sentiment information is not extracted more precisely. Without one of the two layers, the model can only obtain multi-layer de-weighted distribution features or single-layer weight features, which cannot achieve the best performance of multi-layer feature fusion. However, the results did not deteriorate more seriously. We analyze that this may be due to the fact that the number of Conv1D and self-attention in UFEN is set to 2. The increase in the number of processing can make up for the performance degradation caused by the lack of multi-layer feature weight.

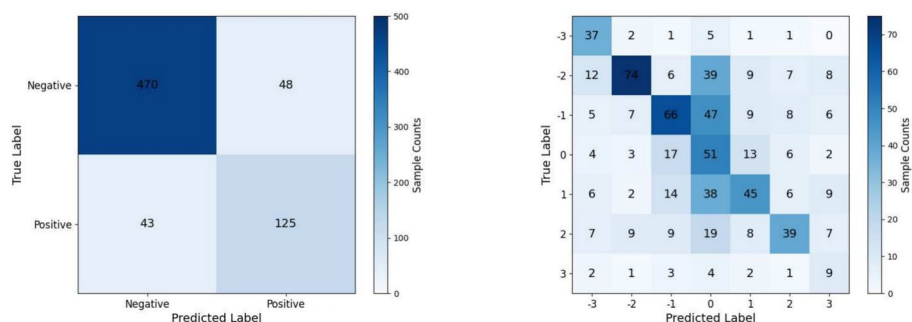
(2) Role of Bi-GRU

Without Bi-GRU, the performance of the model drops significantly. This is because Bi-GRU plays the role of considering both the forward and backward information of the input sequence. Using Bi-GRU can enable the model to understand the context and dependencies in the sequence, which plays an important role in single-modal temporal feature extraction. After removing Bi-GRU, the model cannot accurately represent the coarse-grained sentiment of unimodal data, which are crucial for multi-modal sentiment analysis.

Ablation on MTFN

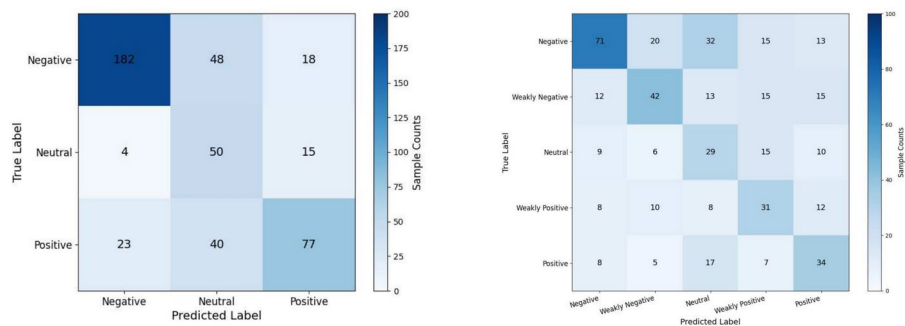
(1) Role of cross-modal attention and encoder-decoder layer

After removing the cross-modal attention and encoder-decoder layers respectively, the performance of the model has declined to a certain extent, especially on more complex task indicators such as Acc5 and Acc7. We analyze that this may be because these two layers focus on high-dimensional feature extraction after modal fusion. Multipolar sentiment analysis requires more information connections between modalities and deep features to maintain stable accuracy. For simple tasks, UFEN has already obtained single-modal sentiment representation. If we ignore the information interaction between modalities and simply fuse single-modal information, the model can also obtain relatively stable Acc2 and corresponding F1-score and Correlation.



(a) Confusion matrix of the two-classification result. **(b)** Confusion matrix of the seven-classification result.

Fig. 6. The confusion matrix of the proposed model's classification results on the MOSI test set.



(a) Confusion matrix of the three-classification result. (b) Confusion matrix of the five-classification result.

Fig. 7. The confusion matrix of the proposed model’s classification results on the SIMS test set.

Paper	Research	Method	Year of publication
Meena et al. ⁵⁶	FER	DCNN	2023
Allenspach et al. ⁵⁷	Drug design	Neural multitask learning	2023
Zhan et al. ⁵⁸	Driving perception	Multitask learning; anchor-free perception	2024
Li et al. ⁵⁹	Equipment fault diagnosis	Multi-domain information fusion	2024
Yu et al. ⁶⁰	Microbe-disease prediction	Multi-kernel fusion	2024
Our method	MSA	CNN; multitask learning; multi-layer feature fusion	–

Table 6. The detailed statistics of SIMS, MOSI and MOSEI datasets.

No. of UFEN	SIMS				MOSI			
	Acc2	F1-score	Corr	Acc5	Acc2	F1-score	Corr	Acc7
1	73.65	73.92	0.530	39.29	83.5/86.0	83.6/86.0	0.781	45.8
2	80.56	80.27	0.583	45.20	85.2/86.6	85.2/86.7	0.792	46.7
3	77.02	76.85	0.563	43.23	83.1/85.2	82.9/85.1	0.782	46.2
4	75.19	75.66	0.527	41.70	81.3/82.8	81.4/82.8	0.756	43.9

Table 7. The impact of different UFEN layer numbers. Significant values are in bold.

(2) Role of multi task learning

Without multi task learning, the performance of the model drops significantly, which indicates that single-task sentiment analysis using fused features after intra-modal and inter-modal information processing cannot achieve the expected results. We analyze that this may be due to the fact that after complex processing, although the fused features have higher-dimensional abstract features, they contain insufficient information. For this situation, adding single-modal sentiment analysis into the model as sub-tasks can solve this problem. At the same time, the sentiment information contained in different modal data is highly correlated, which is in line with the design concept of multi-task learning. Multi-task learning plays an important role in our model. By introducing multi-task learning, the model can well combine multi-modal complementary features and achieve significant improvements in various evaluation indicators.

In order to better understand the impact of hyperparameters on model performance, we conducted ablation studies on learning rate, batch size, training epochs and early stopping batch, as shown in Table 10.

As shown in the table, the model performs optimally with learning rate of 5e–3. Larger learning rates lead to instability during training, preventing the model from converging properly. Smaller learning rates can cause the model to become trapped in local optima. As for batch size, smaller batch size achieved the best results on the SIMS dataset, while a larger batch size was required for the MOSI dataset. We hypothesize this is due to the better data quality of SIMS, which has more detailed sentiment features. This phenomenon is also reflected in the training epochs. SIMS achieves better results in more epochs, but as the number of epochs increases, the accuracy does not continue to increase, and the computational consumption will increase greatly. For early stop batch, overly large parameters prevent the model from fully converging, resulting in a slight decrease in accuracy. Thus, selecting an appropriate early stop parameter can also help reduce computational costs during training.

Visualization analysis

In this section, we conduct visual analysis in order to observe the performance of the model during training and testing more clearly. First, we used t-SNE⁶¹ to visualize the multimodal representation of the MOSI and SIMS

Task Type	SIMS				MOSI			
	Acc2	F1-score	Corr	Acc5	Acc2	F1-score	Corr	Acc7
M	71.15	72.21	0.504	36.57	79.1/81.8	79.0/81.8	0.737	41.5
M, A	72.33	72.99	0.513	39.07	80.0/82.1	80.1/82.1	0.743	42.4
M, T	73.09	73.67	0.527	41.14	80.7/82.4	80.7/82.5	0.754	43.3
M, V	75.49	75.72	0.532	40.29	81.2/83.5	81.1/83.5	0.752	41.8
M, A, T	74.18	74.28	0.529	41.17	82.2/83.9	82.0/83.8	0.761	44.4
M, A, V	76.12	76.31	0.541	41.82	82.6/84.7	82.5/84.6	0.769	42.6
M, T, V	78.96	79.15	0.571	43.26	84.1/86.0	83.9/85.9	0.782	46.3
M, A, T, V	80.56	80.27	0.583	45.20	85.2/86.6	85.2/86.7	0.792	46.7

Table 8. Experimental Results for multimodal sentiment analysis with different tasks using our method. M, A, T, V respectively represent the multimodal, audio, text and vision task. Significant values are in bold.

Model Design	SIMS				MOSI			
	Acc2	F1-score	Corr	Acc5	Acc2	F1-score	Corr	Acc7
(1) UFEN								
w/o Self Att	76.71	77.07	0.575	43.54	82.4/83.8	82.4/83.9	0.758	42.0
w/o Conv1D	76.93	77.11	0.561	42.14	81.9/83.7	81.9/83.7	0.767	45.4
w/o Bi-GRU	73.87	74.12	0.541	38.20	80.5/81.9	80.3/81.9	0.692	39.7
(2) MTFN								
w/o Cross-modal Att	78.34	78.78	0.577	43.33	81.2/83.7	81.1/83.7	0.777	43.6
w/o E-D layer	77.90	75.37	0.559	40.04	82.1/84.5	81.9/84.4	0.782	43.2
w/o Multi task learning	74.74	75.07	0.553	38.39	80.2/81.6	80.0/81.4	0.686	38.5
Complete model	80.56	80.27	0.583	45.20	85.2/86.6	85.2/86.7	0.792	46.7

Table 9. Ablation studies on CH-SIMS and CMU-MOSI datasets. Significant values are in bold.

datasets in the embedding space, the results are shown in Fig. 8. The t-SNE algorithm can map the complex features learned by the model onto two-dimensional feature points, and the model performance is expressed as the clustering effect of different feature categories.

As can be seen from Fig. 8, the model has relatively good classification results in the MOSI and SIMS test sets, which shows that the model effectively captures sentiment-related features and fully explores the information association within and between modalities. In addition, the visualization results show that the intra-class distance is smaller and the inter-class distance is larger in MOSI, while the distribution is more dispersed in SIMS. This phenomenon is similar to the experimental results in “Ablation Study” section, the evaluation index results of the model on SIMS dataset are lower than those on MOSI dataset.

The loss value is an important indicator to quantify the performance of the model. We track the loss during the training process in the training set and validation set, the results are shown in the Fig. 9. All the value of loss demonstrates a decreasing trend with the number of epochs, although the performance on the validation set is not as ideal as on the training set. This phenomenon indicates that the model learned the sentiment representation in the data as we expected.

In order to express the effect of the model more clearly, we select four samples in the SIMS test set to conduct detailed analysis of sentiment analysis results of each modality. The sentiment expressed by these four samples include positive, negative and asymmetric (positive and negative distribution in different modes). The reason for selecting samples in the SIMS dataset is that each modality has a corresponding label, which facilitates researchers to discuss results from each modality. During the experiment, we chose Self-mm as our comparison model because there is not much difference between it and our method in various evaluation indicators. The results are shown in Fig. 10.

For the sample that represent positive sentiment, both methods predict relatively accurately in the fusion mode, but our method is closer to the label. In terms of single-modal sentiment analysis, the comparison model predicts the visual modality, audio modality and text modality of the sample into negative label. Our method accurately predicts visual modality, textual modality as positive. Although our method mispredicts the speech modality as negative, the prediction score is closer to the ground truth than the comparison model. This shows that our model can accurately extract the same type of sentiment information in different modalities and effectively fuse it.

For the sample that represent negative sentiment, both methods made accurate predictions in all modalities. But our method is closer to ground truth in all modalities. This shows that the model is sensitive to negative sentiment features and can identify negative samples more accurately.

For more difficult sample, which is represented as neutral in the fusion mode and audio mode, as positive in the visual mode, and as negative in the text mode. Our model performs better than contrasting methods that

Model design	SIMS		MOSI	
	Acc2	Acc5	Acc2	Acc7
(1) Learning rate				
1e-3/5e-3/1e-4	78.36/ 80.56 /79.60	42.54/ 45.20 /45.00	82.4/ 86.6 /86.1	42.4/ 46.7 /46.3
(2) Batch size				
16/32/64	80.56 /78.74/74.39	45.20 /41.55/36.27	86.2/ 86.6 /83.2	46.2/ 46.7 /42.1
(3) Epoch				
30/50/100	77.29/ 80.56 /79.83	43.35/ 45.20 /44.83	86.6 /86.3/86.3	46.7 /46.4/46.4
(4) Early stop				
4/8/10	80.56 / 80.56 /78.91	45.20 / 45.20 /43.91	86.6 / 86.6 /85.9	46.7 / 46.7 /45.9

Table 10. Ablation studies of hyperparameter on CH-SIMS and CMU-MOSI dataset. Significant values are in bold.

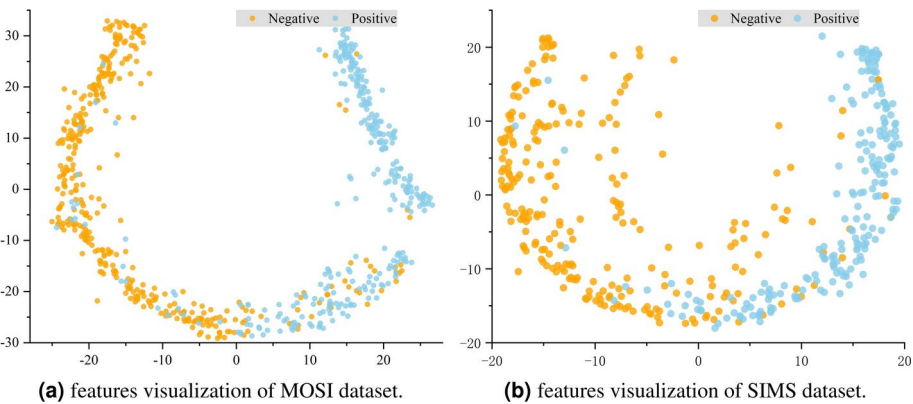


Fig. 8. T-SNE visualization of multimodal representation in the embedding space on the testing set of MOSI (a) and SIMS (b) datasets, where “blue” and “orange” dot denotes positive and negative sentiment respectively.

predict all modalities as negative. Although the fusion modality and audio modality were predicted to be biased toward negative sentiment, the scores were very close to neutral sentiment. This shows that our method can combine the intra-modal and inter-modal information in the sample, and utilize multi-layer feature extraction and inter-modal information interaction to obtain accurate prediction results under unbalanced representations.

In this experiment, we also found that for many samples, both methods predicted the speech modality as negative. We analyze that this may be due to the imbalance of samples in the dataset. In SIMS dataset, negative and neutral sentiment representations account for about 70%, which makes it difficult for model to accurately predict audio modality that contain less sentiment information. This is the problem we would like to research and address in the future.

Conclusion

Accurate multi-modal sentiment analysis requires a model that not only has the ability to extract single-modal sentiment information, but also mines complementary features between different modalities. In this regard, we have proposed a model based on multi-layer feature extraction and multi-task learning methods, including UFEN and MTFN. The UFEN is used to extract multi-layer information and weight coefficients of single modalities. This method makes full use of contextual information while takes into account both global and local sentiment representations, and reduce redundant information interference. The MTFN is used to extract complementary sentiment information and joint representations between modalities. The model uses self-attention and cross-modal attention to deeply explore the interaction of human sentiment information in different modalities and different dimensions. The model parameters are optimized through the prediction results of multiple subtasks, so that the model can obtain stable analysis results even when facing difficult samples with Imbalance representation. Experimental results on MOSI, MOSEI and SIMS datasets show that our model outperforms current state-of-the-art baselines.

After analysis and discussion, for the algorithm model, although the model proposed in this paper has shown certain advantages in the MSA task based on multiple datasets. However, there are still challenges for improvement in achieving more realistic sentiment perception and embodied intelligence. The performance is not stable enough in scenarios with random modal interference and low signal-to-noise ratio. We believe that the proposed algorithm can be improved towards transfer learning and semi-supervised learning to adapt to more complex and dynamic sentiment analysis tasks in real scenarios, achieving stronger generalization performance

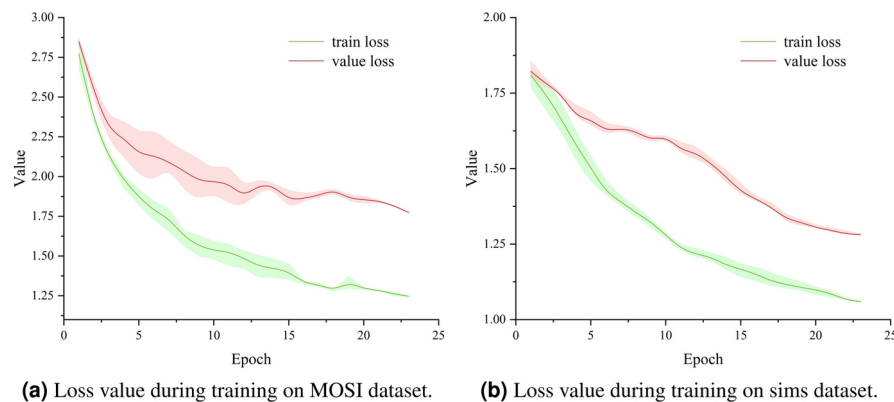


Fig. 9. Trends in the loss values as training proceeds in MOSI (a) and SIMS (b) dataset (five runs across random seeds).

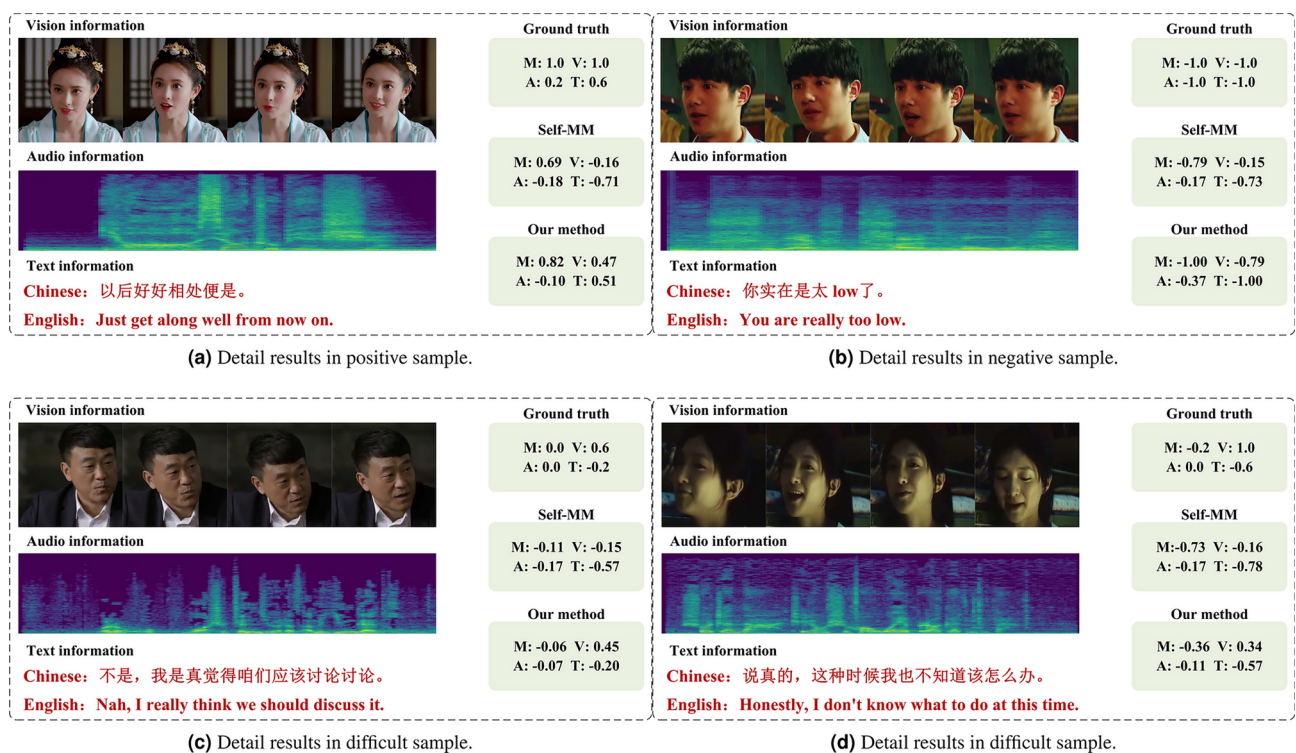


Fig. 10. Detail results of Self-MM versus our method in four test samples of SIMS dataset. (a) Positive sample, (b) is a negative sample, (c,d) are difficult samples, which modalities contain both positive and negative sentiment.

and continuous accuracy improvement. For the dataset, it is our opinion that the current challenges can be divided into two aspects. The first challenge is that sentiment analysis datasets used for specific situations are limited, such as depression detection and anxiety detection. These datasets typically require a large number of volunteers with specific medical conditions and more refined signals, such as ECG, EEG, and EOG. The second challenge is that datasets for general daily HCI struggle to balance the subject's natural state and sentiment masking. Traditional methods for collecting physiological signals like EMG and ECG can cause discomfort for the wearer, while signals such as facial expressions and voice are often subject to deliberate masking. Combining visual sensors with radio frequency sensors is a promising solution, enabling joint non-contact sensing of facial expression signals and physiological signals.

This work only focuses on the multimodal sentiment information that from the most common visual, audio and text signals, does not include more modal information. In future work, we plan to introduce physiological signals collected by hyperspectral cameras or millimeter-wave radars into multimodal sentiment analysis. These sensors can also ensure non-contact remote information collection, which is in line with the current

development trend of human-computer interaction. At the same time, the introduction of physiological signals can better solve the problems such as modal loss and sentiment deception that currently face.

Data availability

All data supporting this research are from previously reported studies and datasets, which have been cited within the article. The MOSI dataset is available for download through <http://multicomp.cs.cmu.edu/resources/cmu-mosi-dataset/>. The MOSEI dataset is available for download through <http://multicomp.cs.cmu.edu/resources/cmu-mosei-dataset/>. The SIMS dataset is available for download through <https://github.com/thuiar/MMSA>.

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Declarations

Competing interests

The authors declare no competing interests.

Additional information

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